

2D SCANNING LIDAR · LR-1BS SERIES

25-METRE SLAM OLEI LR-1BS5H 270° LIDAR

A 270° scanner built for mapping rather than zone relays: 25 metres of usable range, $< \pm 20$ mm absolute accuracy on common targets, a user-settable weak-signal threshold and multi-level filtering to clear tailing artefacts before they reach your SLAM stack.



■ WHY THIS SENSOR

- **0.2–25 m at 80% reflectivity** — two and a half times the LR-1BS2 envelope — long enough to map a warehouse aisle end to end
- **$< \pm 20$ mm absolute accuracy on common targets** — $< \pm 40$ mm on reflectors, ≤ 20 mm repeat — tight enough for localisation, not just avoidance
- **User-settable weak-signal threshold** — raise it in a dusty bay, lower it for dark targets, without changing the mount
- **Built-in multi-level filtering** — configurable tailing suppression at the sensor, so the point cloud arrives clean
- **40K points per second, 0.225° fixed resolution** — azimuth, distance and reflection intensity per point over 100 Mbps Ethernet
- **$\leq \pm 1^\circ$ horizontal plane error** — a scan plane you can trust when the map depends on it
- **M12 power and LAN connectors, IP65, 450 g** — 5-pin M12 A-code power and 4-pin M12 D-code LAN, both on 0.25 m pigtailed
- **OLAMViewer and a ROS driver** — drop it into an existing SLAM pipeline without writing a parser

0.2–25 m

WORKING RANGE

270°

FIELD OF VIEW

$< \pm 20$ mm

ABSOLUTE ACCURACY

40K

POINTS / SECOND

■ APPLICATION SCENARIOS



SLAM NAVIGATION

Point-cloud mapping and localisation for mobile platforms



AMR & AGV

270° obstacle avoidance from a single chassis-mounted scanner



DOCKING & TRACKING

Reflector-based docking and pallet or object tracking



INTRALOGISTICS

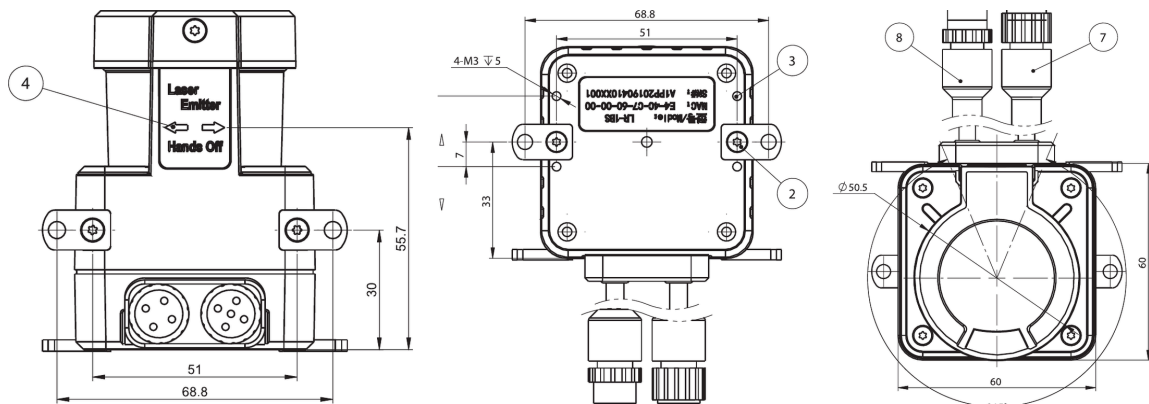
Forklifts, tuggers and conveyor zones in active aisles

LR-1BS5H — SPECIFICATIONS

PARAMETER	TECHNICAL DATA
Sensing technology	2D scanning ToF LiDAR
Laser wavelength	905 ± 20 nm
Laser safety level	Class 1 (IEC 60825-1:2014)
Laser channels	1
Field of view (horizontal)	270°
Beam divergence	5.8 mrad longitudinal 0.58 mrad transversal
Angle resolution (horizontal)	0.225°
Scanning frequency	10–25 Hz
Measurement range (based on reflectivity)	0.2–25 m @ 80% reflectivity 0.2–8 m @ 10% · 0.2–2 m @ 4% · 0.2–25 m @ reflector
Accuracy ①	Common targets: absolute < ±20 mm Reflector: absolute < ±40 mm · Repeat ≤ 20 mm
Distance resolution	1 mm
Signal strength	0–60000
Scanning rate	16K–40K pts/s
Horizontal error (horizontal angle)	≤ ±1°

PARAMETER	TECHNICAL DATA
Switching outputs	1
Ethernet connection	100 Mbps Ethernet
Connection	5-pin M12 A-code male, 0.25 m power 4-pin M12 D-code female, 0.25 m LAN
Operating voltage	12–24 V DC
Power consumption @ 25 °C	5 W
Mechanical dimensions	60 × 60 × 85 mm
Weight	450 g
Degree of protection	IP65 or optional
Operating temperature	–10 to +50 °C
Storage temperature	–30 to +70 °C
Air humidity	< 95% RH
Ambient light limit	80 000 lux
Indicator status	RGB × 4
Mounting	Bottom / back
Basic software	OLAMViewer · ROS driver

OUTLINE AND MOUNTING



UNIT: MM · DRAWINGS NOT TO SCALE

Certification. FCC, EMC, CE, RoHS, REACH. EMC per IEC 61000-6-2:2016-08 and IEC 61000-6-3:2006-07. Vibration per IEC 60068-2-6:2007. Shock per IEC 60068-2-27:2008.

Functional safety. The manufacturer's specification for this product states laser safety Class 1 per IEC 60825-1:2014 and the EMC, vibration and shock standards listed above. It does not state an IEC 61496 Type rating or an ISO 13849 performance level. Treat the zone outputs as obstacle-avoidance and area-monitoring signals, not as a rated safety function, and do not use this sensor as the sole protective device. Where a rated scanner is required, ask MorpheusTEK about the OLEI GS1-5.

Notes. ① Standard value; actual value depends on environmental conditions. The 25 m figure is the 80% reflectivity and reflector case; plan on 8 m at 10% reflectivity and 2 m at 4%. Weak-signal threshold and filter level are user settings — record the values used when a map is commissioned.

Installation. Mount the scanner so the 270° plane is unobstructed through its full sweep and the blind 90° faces structure, not traffic. Keep the optical window clear of spray and dust and clean it on a schedule in dirty environments. Re-run the map after any change to mounting height or angle.

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